Alignment and Fusion in Biological Foundation Models

Literature Review

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Introduction

Biological prediction can draw on molecular sequences, structures, and measurements of cellular states. These forms of information describe different aspects of biology and support different prediction tasks. Biological foundation models provide learned representations that can be adapted to tasks such as predicting transcript abundance or RNA structure (Garau-Luis et al., sec. 4; Yang et al. 21930–33). Comparing such models requires considering what information they receive, how they represent it, and what their evaluations establish about the resulting predictions.

Combining biological representations introduces a further question about the relationships between them. In the studies considered here, alignment can describe positional correspondence imposed before fusion, relationships encouraged through training, or similarity measured between learned representations (Mollaysa et al., sec. 2.1; Chen et al., sec. 3.1; He et al. 950, supplementary note 6). These approaches establish different kinds of correspondence, whose usefulness must be assessed for the task being performed. This review therefore asks when a specified relationship between biological representations helps a defined biological task. It compares how studies construct or measure that relationship and what evidence supports its biological interpretation and predictive value.

The discussion begins with the biological information contained in model inputs and the meaning of prediction targets, followed by the computational roles of representations, attention, and training. It then compares fusion approaches and joint learning before examining the different relationships described as alignment. Table 1 brings together the studies' roles, approaches, findings, and limitations. The evaluation discussion considers what model

comparisons and benchmarks establish, and the final Summary connects these findings to the choice of a method for a particular biological task.

From Biological Sequences to Model Predictions

DNA and RNA are chains of nucleotides; proteins consist of one or more chains of amino acids. Transcription uses DNA as a template for RNA. During translation, the ribosome reads messenger RNA (mRNA) in three-nucleotide codons, while transfer RNAs (tRNAs) supply the corresponding amino acids. The ribosome joins these amino acids into a polypeptide chain; the mRNA supplies instructions rather than becoming part of that chain. Nucleotide and protein sequences therefore describe related biological objects (Alberts et al., *Essential Cell Biology*, ch. 7, “From DNA to RNA” and “From RNA to Protein”). In comparing their models, a modality means a form or channel of information: RNA sequence and RNA structure can be distinct modalities even though both describe RNA. A biological property such as RNA stability may instead be the quantity predicted from those inputs. These roles depend on how a study constructs its task.

This relationship leaves an important question about model inputs: which part of the DNA or RNA sequence is supplied? A gene occupies a region of DNA, but information affecting its expression can also lie outside its protein-coding sequence. Non-coding regulatory DNA, for example, helps control transcription (Alberts et al., *Molecular Biology of the Cell* 334, 415). To include surrounding context, a model may receive a selected interval of genomic DNA, called a genomic window. IsoFormer uses such an interval around the location where transcription begins. The window’s boundaries define the model’s input; they do not imply that the entire interval is transcribed (Garau-Luis et al., sec. 4.4).

The RNA produced from a protein-coding eukaryotic gene undergoes processing as it matures. For transcripts containing introns, splicing removes those regions and joins the retained regions, called exons. Alternative splicing changes which regions are retained, allowing one gene to produce different transcript forms. Exons are not necessarily entirely protein-coding: they can also contain untranslated sequence. Thus, the mature transcript differs both from the genomic region that produced it and from the portion that specifies a protein (Alberts et al., *Molecular Biology of the Cell* 195, 338–41, 446–47).

[Biological clarification: Explain introns and exons in more detail, with an example showing which regions are removed or retained during splicing and how retained regions can include untranslated sequence.]

Within that mature mRNA, the coding sequence is the portion read during translation to specify the protein. The untranslated regions lie on either side of the coding sequence and remain in the mature mRNA without being translated into that protein. They can influence translation, localization, and decay. Localization describes the mRNA's location within the cell, while decay is its degradation over time. Including the full transcript can therefore provide information absent from the coding sequence alone or its translated protein sequence (Alberts et al., *Molecular Biology of the Cell* 455–56, 460).

[Biological clarification: Clarify the processing criteria for calling this transcript mature mRNA, including how intron retention (IR) affects that interpretation, and distinguish RNA maturation from protein maturation.]

These differences in sequence content must be distinguished from changes in how the same sequence is represented. BioLangFusion formulates its inputs as corresponding DNA, mRNA, and protein sequences. At the level of sequence notation, replacing U with T can

produce a DNA-alphabet version for a nucleotide encoder; it does not reproduce biological transcription or recover introns and surrounding regulatory DNA. By comparison, IsoFormer's genomic window supplies context beyond its transcript input. The relevant distinction is what sequence information becomes available, rather than whether the inputs have been assigned different molecular labels (Mollaysa et al., sec. 2; Garau-Luis et al., sec. 4.4).

Deriving a protein sequence requires identifying the coding region and its reading frame, which specifies where to begin grouping consecutive nucleotides into triplets—codons. The genetic code then maps each codon to an amino acid or a stop signal. Because several codons can specify the same amino acid, translation loses the distinction between those synonymous codons; the protein sequence generally cannot identify the original coding sequence uniquely (Nelson et al., sec. 27.1). This explains how a derived protein input can retain amino-acid identity while omitting nucleotide-level distinctions present in the original input.

Models must represent these biological sequences in a form that numerical operations can process. An encoder converts sequence inputs into numerical representations—embeddings. In DNABERT, the input is first divided into tokens consisting of overlapping k-mers—groups of k nitrogenous bases. ATGGCT yields ATG, TGG, GGC, and GCT for k = 3, whereas reading codons from the first base gives ATG and GCT. Each token receives an initial vector, a list of numbers, which transformer layers update using information from other positions. The resulting embeddings depend on sequence context. The term encoder describes the component's role, while transformer describes its architecture. Tokenization therefore need not follow the biological grouping used in translation (Ji et al. 2113–14, fig. 1).

Choosing an encoder introduces a distinction beyond the biological information loss described above. Translating a coding sequence changes which molecular features are explicitly

represented, while the encoder changes how the supplied sequence is transformed. A protein encoder and a nucleotide encoder can consequently make different patterns accessible to a later predictor, even when their inputs originate from the same transcript. Whether combining their representations helps a particular prediction requires an experimental comparison (Mollaysa et al., secs. 2–3).

A prediction head then maps the model's representations to a requested quantity or category. Some tasks require a summary of a whole sequence, so pooling combines representations across positions, for example by averaging them. IsoFormer uses such a summary to predict transcript abundance across tissues. Here, the embeddings supply inputs to the predictor, while measured transcript abundance provides reference values for training and evaluation (Garau-Luis et al., secs. 4.3–4.4). Interpreting the prediction therefore requires understanding what abundance measures and how it differs from a related target such as RNA stability.

The target determines what a successful prediction means biologically. RNA abundance describes how much RNA is present under specified conditions: synthesis adds molecules and degradation removes them. RNA stability instead concerns how long molecules persist, often described by their half-life. Consequently, an abundance difference can arise from changes in production or degradation, whereas a stability measurement addresses persistence more directly. IsoFormer's abundance predictions and BioLangFusion's stability predictions concern related but different biological quantities (Nelson et al., sec. 26.2, "Cellular mRNAs Are Degraded at Different Rates"; Garau-Luis et al., sec. 4.4; Mollaysa et al., sec. 3).

Molecular measurements can also serve as model inputs. MicroRNAs (miRNAs) are short noncoding RNAs that help reduce target-mRNA translation and promote degradation. DNA

methylation adds methyl groups to DNA bases and can contribute to gene regulation, although a methylated position alone does not establish gene inactivity (Alberts et al., *Molecular Biology of the Cell* 435–36, 463–64). Coupled Mamba combines measurements of mRNA and miRNA abundance with DNA methylation for breast-cancer subtype classification, using molecular states rather than sequences (Li et al., sec. 4.1, table 5).

Other tasks concern molecular structure rather than RNA quantity. A polypeptide chain can fold into a three-dimensional arrangement—its conformation—and folding can begin while the chain is being constructed at the ribosome. This arrangement can bring amino-acid residues far apart along the sequence close together in space (Alberts et al., *Molecular Biology of the Cell* 380; Nelson et al., sec. 4.4). RNA also folds: complementary regions within an RNA molecule can pair, forming features such as stems and loops (Nelson et al., sec. 8.2). A structural label is an annotation describing such a feature. RNA secondary-structure labels can specify paired and unpaired positions. For the contact task evaluated in Structure-Aligned Protein Language Model, two residues count as being in contact when their alpha-carbon atoms are within 8 Å (0.8 nanometers) of one another. This distance rule defines a prediction target; it does not provide a complete measurement of molecular behavior inside a cell (Yang et al. 21930–33; Chen et al., sec. 4.3.1).

[Conditional extension: Explain how chaperones and cellular conditions influence protein folding if later discussion connects sequence or structure predictions to protein function.]

Genomic context, mature transcripts, and translated proteins therefore provide related but noninterchangeable information: transcript processing changes the sequence context, and translation removes some nucleotide-level distinctions. Their value in combination remains an empirical question tied to a biological target; predicting RNA abundance differs from predicting

RNA stability. Assessing alignment consequently requires following both the information preserved in each input and the meaning of the prediction. The next section explains how encoders represent those inputs, fusion combines their representations, and training adapts the system to its task.

Representations, Attention, and Training

In the systems compared here, sequence encoders produce representations, a fusion component combines information from those representations, and a prediction head maps the result to the biological target. A transformer can provide the encoder architecture within this larger system. Following this path distinguishes the units being represented, the information exchanged, and the components that learn during training (Mollaysa et al., sec. 2; Garau-Luis et al., secs. 4.1–4.4).

The embeddings introduced above have two dimensions relevant to fusion: the number of represented positions and the number of features in each vector. A position may represent a nucleotide token, k-mer, or amino-acid residue. Tokenization affects the first dimension; encoder architecture determines the second. Combining streams position by position requires equal position counts and a meaningful biological basis for pairing them. For example, a supplied coding sequence and reading frame identify which nucleotide triplet corresponds to each protein residue. Learned projections can supply a common feature width where needed. Matching these dimensions does not demonstrate that the representations encode the same biological properties (Mollaysa et al., secs. 2.1–2.2).

Self-attention updates token representations using information from positions within the same sequence, including their own. DNABERT's transformer encoder implements this with learned projection matrices that produce query, key, and value vectors. A query from one

position is compared with keys from the available positions; the resulting scores become weights for combining their value vectors. Different attention heads use different projections, and successive layers update the representations using the context accumulated so far. Thus, the representation of a base group depends on the sequence around it rather than only on its letters (Ji et al. 2113–14, fig. 1).

[Worked computational example: Add a worked numerical example of queries, keys, and values to illustrate how attention combines token representations.]

Cross-attention extends this operation to another source of representations. The stream being updated supplies queries; another stream, or a combined set of streams, supplies keys and values. The streams need not have equal token counts. The distinction from self-attention concerns which positions can contribute information to a representation, rather than whether the model's learned parameters change (Mollaysa et al., sec. 2.2.3; Garau-Luis et al., fig. 1, sec. 4.2).

The resulting attention weights also require careful interpretation. DNABERT's Figure 4 examines contextual attention around biologically relevant sequence regions and discusses possible model bias. Such patterns can suggest regions for further investigation, but an attention map does not establish a causal regulatory mechanism. This within-sequence analysis also differs from evaluating alignment between representations produced for different modalities (Ji et al. 2116).

Pretraining is an earlier stage of training that learns parameters, including weights and biases, for reuse on later tasks. In a transformer encoder, those parameters are used to calculate contextual embeddings—representations that incorporate sequence context—and attention weights for each supplied input. Different inputs can therefore produce different representations

and attention weights even when the learned parameters remain unchanged (Ji et al. 2113–14, fig. 1).

An encoder can remain frozen while fusion and prediction components learn from its outputs, or undergo fine-tuning so its pretrained parameters also change (Mollaysa et al., secs. 2.2–4; Garau-Luis et al., secs. 4.1, 4.4). These alternatives determine which parts adapt to a biological target. Interpreting fusion requires examining both the information exchanged through attention and the parameters changed during training.

Comparing Fusion Approaches

Fusion in BioLangFusion and IsoFormer

BioLangFusion and IsoFormer apply these operations to different biological questions. BioLangFusion first maps sequence representations to codon-level resolution under an assumed coding correspondence, then compares three fusion methods. IsoFormer combines genomic context, a transcript, and its associated protein to predict transcript abundance. Their input mapping, prediction heads, and training choices clarify what each evaluation can establish (Mollaysa et al., secs. 2–3; Garau-Luis et al., secs. 4.1–4.4).

BioLangFusion's alignment scheme assumes corresponding coding sequences and uses three encoders: Nucleotide Transformer v2 supplies DNA representations at six-base positions, RNA Foundation Model (RNA-FM) supplies RNA representations at individual-nucleotide positions, and ESM-2 supplies protein representations at amino-acid positions. These different token sizes yield different numbers of contextual embedding vectors for the corresponding sequences (Mollaysa et al., sec. 2.1, table 3).

BioLangFusion uses the number of amino-acid positions in the protein representation as the reference length for all three streams. Under this coding-sequence assumption, the reading

frame identifies the codon corresponding to each residue. The DNA and RNA representations are then adjusted to the same length—one position for each residue and its matched codon. For DNA, a learned transposed convolution increases the number of embedding positions; Appendix A.4 also specifies padding to the protein-reference length. For RNA, a separate mean-pooling operation averages the three vectors associated with each consecutive, nonoverlapping nucleotide triplet. This produces representations at codon-level resolution without demonstrating that the expansion of the DNA representation has recovered distinct biological information for each codon (Mollaysa et al., sec. 2.1, eq. 1, app. A.4). Untranslated regions and arbitrary genomic windows do not support the same direct pairing with protein residues (Nelson et al., sec. 27.1). After alignment, BioLangFusion compares three fusion methods: concatenation, Multiple Instance Learning (MIL) with Entropy Regularizer, and cross-modal multi-head attention (Mollaysa et al., sec. 2.2).

Concatenation joins feature vectors at corresponding positions side by side. BioLangFusion first reduces the DNA vectors' width with a learned projection, then appends the RNA and protein features at each codon position. The learned projection changes the features within each DNA vector without changing the number of represented positions. The prediction model can learn relationships among these combined features, but concatenation itself does not calculate a weight for each modality from the input. It therefore provides a baseline for asking whether attention-based weighting or interactions improve prediction beyond jointly supplying the features (Mollaysa et al., sec. 2.2.1).

The MIL + Entropy method adds such weighting. It maps all three streams to a common feature width and uses a mean-pooled summary of each stream to calculate one weight per modality for each input sequence. Those weights combine the representations at their

corresponding codon positions. The summary used to calculate a weight is therefore different from the sequence of fused representations passed onward. An entropy penalty encourages less uniform weights, allowing the model to emphasize selected streams. This describes how the predictor allocates information; it does not show that the favored modality is the biological cause of the measured outcome (Mollaysa et al., sec. 2.2.2, fig. 2).

The third method, cross-attention, allows a position in one stream to draw information from positions across all three streams. After projection to a common feature width, the three sequences of vectors are stacked into a shared context. This joins them along the sequence length axis—the dimension that counts represented positions—rather than widening the vector at each position. Each modality queries that context, so an update can use other codon positions as well as the corresponding one. The updated streams are then combined into a sequence of fused vectors (Mollaysa et al., sec. 2.2.3, eqs. 3–5).

Each fusion method passes its combined vectors to a TextCNN prediction head—a convolutional network that converts these features into task predictions. Its filters detect local patterns across neighboring fused positions, and global max pooling retains the strongest response across positions in each learned feature channel. A fully connected output layer then produces the task prediction, such as a numerical RNA-stability score. Fusion supplies the combined representations; the TextCNN prediction head converts them into that score (Mollaysa et al., sec. 3, app. A.4, lists 1–2).

On RNA stability, BioLangFusion's gains depend on the fusion method, and inconsistent descriptions of the prediction head limit attribution of the differences to fusion alone. The detailed results and this qualification are examined in the evaluation section (Mollaysa et al., sec. 3, table 1, app. A.4).

IsoFormer predicts the abundance of individual RNA transcript isoforms—distinct transcript forms produced from the same gene. This target differs from total gene expression, which combines the abundances of that gene's transcripts (Garau-Luis et al., sec. 4.2, eq. 4). The study uses bulk RNA-sequencing measurements from GTEx, reported as transcripts per million (TPM), a normalized measure of relative abundance. It averages each transcript's abundance across individuals within each of thirty tissues. For training, these values undergo a log(1 + value) transformation followed by normalization within each tissue. The resulting target describes a transcript's average abundance in a tissue (Garau-Luis et al., sec. 4.4, apps. A–B).

To predict this abundance, IsoFormer supplies the encoders with a DNA region centered on the transcript's transcription start site, the transcript sequence, and its associated protein sequence when the transcript is protein-coding. The DNA region provides surrounding regulatory context, while the RNA input can include untranslated sequence absent from the protein (Garau-Luis et al., secs. 4.2–4.4). Its initial configuration uses separate Nucleotide Transformer v2 encoders for DNA and RNA (250 million parameters each) and ESM-2 for protein (150 million parameters). Replacing U with T lets the DNA-pretrained Nucleotide Transformer process RNA using its existing vocabulary. The Nucleotide Transformer and ESM-2 checkpoints differ from BioLangFusion's, despite the shared model-family names (Garau-Luis et al., sec. 4.1, table 1; Mollaysa et al., table 3).

IsoFormer combines the resulting embeddings in an aggregation module—the part of this architecture that performs fusion. Each stream successively uses cross-attention to incorporate information from the other two. For example, DNA positions can draw on transcript and protein positions in separate cross-attention steps. A residual connection adds each attention-derived update to the existing representation, keeping the dimensions of the stream being updated. The

updated streams are then combined for prediction. Unlike BioLangFusion's imposed codon mapping, these interactions connect representations of different sequence contexts without requiring codons to be matched with corresponding protein residues (Garau-Luis et al., fig. 1, secs. 4.1–4.2).

The prediction head turns this combined information into one abundance prediction per tissue. It uses global average pooling to summarize representations across positions, followed by a linear output layer. This differs from BioLangFusion's TextCNN prediction head, which first detects local patterns with convolutional filters and then retains their strongest responses through max pooling (Garau-Luis et al., sec. 4.3; Mollaysa et al., app. A.4). IsoFormer also fine-tunes its pretrained encoders jointly with the aggregation module and prediction head. Its mean squared error objective penalizes squared differences between predicted values and the processed expression targets, allowing all three components to adapt to the task. BioLangFusion instead trains the later components using extracted pretrained embeddings (Garau-Luis et al., sec. 4.4, app. B; Mollaysa et al., secs. 2.2–4).

Ablation studies remove or replace inputs or components to assess their contribution. IsoFormer's modality ablation compares configurations that include or omit particular sequence inputs, supporting a benefit from additional inputs under the tested setup. Its aggregation ablation replaces the fusion module with alternatives examined in the evaluation section; it does not establish a unique advantage for cross-attention. These comparisons test different contributions to prediction (Garau-Luis et al., secs. 5.1–5.3, tables 2–4).

These cases distinguish the biological basis for pairing inputs from the operations that exchange information between their representations. BioLangFusion imposes a codon mapping before fusion; IsoFormer's cross-attention connects different sequence contexts. Their prediction

heads and training arrangements also differ, so neither positional matching nor attention alone demonstrates that the encoders have learned similar representations. Interpreting either system requires tracing what information enters, where it can interact, which components learn, and what evaluation tests its value. These questions also apply to approaches that organize information exchange around a biological direction or combine molecular information during pretraining (Mollaysa et al., secs. 2–4; Garau-Luis et al., secs. 4–5).

Directional integration and joint learning

Central Dogma Transformer organizes computational information flow around the DNA-to-RNA-to-protein relationship. Through cross-attention, RNA representations incorporate information from DNA representations; protein representations then incorporate information from the updated RNA representations. As a proof of concept, the study tests the approach on one prediction task using an existing CRISPR-interference (CRISPRi) screen in K562 cells. The screen suppresses candidate enhancers, regulatory DNA regions that can influence gene expression. The model predicts the measured change in gene expression following this intervention (Ota 28–32, secs. 7.1, 7.8).

Although the prediction target measures an expression change, the model does not receive corresponding changes in RNA and protein measurements as inputs. Its RNA input represents gene identities without including expression values, which measure RNA abundance in each example. These gene embeddings and the corresponding protein embeddings are arranged in matrices—tables containing one embedding vector per row. The RNA and protein rows are matched by gene identity, and the same two input matrices are reused while the DNA representation changes with the enhancer being considered. Frozen encoder parameters can still produce different embeddings for different inputs; here the supplied RNA and protein matrices

themselves remain fixed across examples. Cross-attention can subsequently update these representations using the varying DNA information. They provide learned gene and protein context, but predictive success does not itself establish recovery of transcriptional or translational mechanisms (Ota 29–32, secs. 7.2–7.8).

[Biological clarification: Explain how CRISPRi targets regulatory regions and how the screen measures their effects on gene expression.]

Whereas the preceding systems combine outputs from separately pretrained encoders, MIMIC learns relationships across molecular information during multimodal pretraining. It organizes sequence-aligned features into two main tracks: a nucleotide track and an amino-acid track. Codons share the amino-acid track because each supplied codon is paired with its corresponding residue. These track coordinates refer to positions along sequences. Within each track, representations assigned to a position have the same number of numerical entries—a common vector width—so they can be added element by element. Width describes the size of each vector, not the number of sequence positions. Contextual information, such as tissue labels, occupies separate token groups. This arrangement preserves positional correspondence while allowing different kinds of information to enter a shared model (Golkar et al., sec. 2, apps. A.3, B.1–B.2).

The encoder processes the supplied inputs, and the decoder uses cross-attention to draw on the resulting representations when predicting selected missing information. For example, one sequence-completion experiment hides 100 nucleotides in an unspliced transcript and asks the model to reconstruct them using the remaining sequence and auxiliary information. This makes the role of additional inputs concrete: they provide context for recovering the missing sequence. Whether the learned representations also support predictions of RNA properties is a separate

question addressed through mRNABench in the evaluation section (Golkar et al., secs. 4.1–4.2, apps. B.1, D.1, D.3).

Central Dogma Transformer specifies a direction for information exchange, while MIMIC uses corresponding sequence positions to combine features during multimodal pretraining. These choices explain how information is connected within each model. The following studies examine a different question: what relationships between representations are learned or measured, and how those relationships support a specified task.

What alignment establishes

LucaOne reports convergence between embeddings of nucleic-acid and protein sequences associated with the same gene. Its unified model trains on both sequence types without explicitly pairing corresponding nucleotide and protein sequences during pretraining (He et al. 943, 950). LucaOne's masked-sequence prediction tasks recover deliberately hidden nucleotide or amino-acid identities from sequence context (He et al. 949). MIMIC's sequence-completion evaluation similarly reconstructs hidden sequence identities, while testing how additional biological inputs affect reconstruction (Golkar et al., sec. 4.1, app. D.1). LucaOne also learns from biological annotations, including labels identifying coding regions, introns and binding sites within proteins (He et al. 948–49). The convergence analysis asks whether the two sets of embeddings organize other genes similarly. For 1,284 gene–protein pairs, the authors mean-pool final-layer embeddings and identify nearest neighbors separately within each set. These are the closest embedding vectors by Euclidean distance, rather than adjacent positions in a molecular sequence. The authors compare the neighboring gene identities using Jaccard similarity, calculated as the number of shared neighbors divided by the number of distinct neighbors across both sets. They

then average these scores across the gene–protein pairs (He et al., supplementary note 6, eq. 16, and supplementary fig. 12b).

The convergence result therefore concerns similar neighborhood relationships, not equality of paired vectors. LucaOne separately tests whether supplied DNA and protein sequences form a valid pair. BioLangFusion's codon mapping addresses another relationship: an imposed correspondence between sequence positions before fusion. LucaOne's neighborhood comparison and sequence-pair test, together with BioLangFusion's codon mapping, address distinct relationships. None of these relationships alone demonstrates that combining representations improves prediction (He et al. 943, 950; Mollaysa et al., secs. 2–3).

Where LucaOne measures convergence in learned representations, Structure-Aligned Protein Language Model explicitly encourages sequence–structure correspondence during training. At the level of individual amino-acid residues, contrastive learning trains a sequence representation to score more highly with the structure representation of the same residue in the same protein than with representations of other residues in the training batch. A second objective, structure-token prediction, uses discrete labels describing the local three-dimensional geometry around each residue. The sequence encoder learns to predict these labels, which describe aspects of structure rather than complete three-dimensional coordinates. These objectives use structural representations and labels to supervise a sequence encoder; the resulting SaESM2 encoder can work from sequence alone at inference. Contact prediction then tests whether structural information can be recovered from the sequence representations (Chen et al., secs. 2.3, 3.1, 4.3.1).

OmniGenome also connects sequence with structure, but learns prediction in both directions for RNA. Its pretraining includes predicting secondary-structure labels, which describe

paired and unpaired nucleotide positions, from sequence. In the reverse direction, it predicts deliberately hidden nucleotides using the remaining sequence and structural context. The training structures are computational predictions rather than experimentally determined structures, which matters when interpreting what supervision the model receives (Yang et al. 21930–31). Its RNA-design evaluation supplies a desired secondary structure and asks whether the model can help find a sequence predicted to form that base-pairing pattern. The model operates within a genetic algorithm, a search procedure that repeatedly modifies candidate sequences and retains promising candidates (Yang et al. 21933).

MSAlign gives alignment a different task: identifying which candidate molecule corresponds to an experimental mass spectrum. In tandem mass spectrometry, a selected charged molecular ion is broken into charged fragments. The resulting spectrum records their signal intensities and mass-to-charge ratios, each fragment's mass divided by its charge number. MSAlign ranks candidate molecular structures, descriptions of atoms and their bonds, to identify the molecule that produced the measured spectrum (Krzakala et al., secs. 1–2).

The spectrum encoder represents the measurement, while the molecule encoder represents each candidate structure. Both encoders remain frozen, with unchanged parameters; trainable projections map their outputs into a shared representation space. The contrastive objective favors a spectrum paired with its correct molecule over pairings with other molecules. Candidate-based negatives come from that spectrum's candidate list; in-batch negatives come from other examples in the training batch. Comparing these alternatives tests whether training against the candidates encountered during retrieval improves identification. Here, alignment is assessed through its usefulness for identifying the correct molecule among candidates (Krzakala et al., secs. 3, 5.3, table 5).

Beyond performance on a particular design or retrieval task, Understanding the Emergence of Multimodal Representation Alignment asks whether higher measured alignment consistently corresponds to better performance. Its experiments with artificially generated data use unbiased Centered Kernel Alignment (CKA), which assesses whether representations encode similar relationships among examples without requiring their individual coordinates to match. Its experiments with large pretrained vision and language models use mutual k-nearest neighbors (mutual KNN). For each image–caption pair, the analysis finds the k nearest other examples separately in the image-embedding and caption-embedding spaces, then compares the identities appearing in those neighbor sets. The study distinguishes task information available in both modalities from task information available in only one. It finds that higher measured alignment does not consistently correspond to better performance. For biological fusion, this motivates examining whether models preserve useful differences as well as common information, and checking how alignment scores relate to performance on the intended task. That application remains a hypothesis because the synthetic and nonbiological experiments do not validate a DNA–RNA–protein architecture (Tjandrasuwita et al., secs. 2–3.1, 5–6, app. B).

The biological distinctions establish what the inputs describe and which information they make available to a model. Genomic context, processed transcripts, proteins, and structural annotations describe different features; a sequence derived from another does not necessarily supply an independent observation. Computational choices determine how this information becomes available to a predictor. Tokenization establishes represented units, encoders produce numerical features, and fusion combines representations. Alignment can concern positional correspondence, relationships learned between representations, or the matching of related inputs.

These relationships must be distinguished when comparing what methods offer for a biological task.

Table 1 groups studies by their principal roles: models and representation methods; and comparative evaluations, benchmarks, and analyses. These roles overlap because model papers also evaluate their methods. Its columns connect each study's data and research question to its approach, findings, limits, and implications, linking the preceding explanations to the evaluation evidence that follows.

Table 1 Contributions and limits of the selected studies

Study and role	Data and research question	Approach	Findings and limits	Implication for model comparison
Models and representation methods				
DNABERT Unimodal reference	DNA k-mers; prediction from regulatory sequences.	Transformer encoding and contextual attention within a DNA sequence.	Attention patterns can reflect model bias as well as biological features.	Distinguish attention-based interpretation from cross-modal alignment. Ji et al. 2113–14, 2116, fig. 4.
BioLangFusion Sequence fusion	DNA, mRNA, and protein views assumed to share coding correspondence; sequence-property prediction.	Impose codon-level positional correspondence, then compare fusion operations.	Gains vary by task and fusion; descriptions of head settings conflict.	Position mapping alone establishes neither representation similarity nor fusion benefit. Mollaysa et al., secs. 2–3, table 1, app. A.4.
IsoFormer Sequence fusion	Genomic context, transcript, and protein; tissue-abundance prediction.	Cross-attention with joint fine-tuning; compare input, encoder, and fusion choices.	Added inputs improve the tested setup; fusion comparisons do not uniquely favor cross-attention.	Separate changes to available information, encoders, and fusion. Garau-Luis et al., secs. 4.2–4.4, tables 2–4.
Central Dogma Transformer Directional fusion	Varying enhancer DNA with reused RNA/protein input matrices; enhancer-effect prediction.	RNA representations attend to DNA; protein representations then attend to updated RNA.	RNA/protein input matrices remain fixed across examples; predictive accuracy alone does not establish recovery of cellular mechanisms.	Identify which inputs vary before attributing gains to information exchange. Ota 29–32.
LucaOne Unified pretraining	Nucleic-acid and protein sequences; cross-molecule representation relationships.	Unified sequence and annotation pretraining; separate neighborhood and valid-pair evaluations.	Reports neighborhood convergence; the sequence-pair test assesses a separate relationship.	Convergence and valid-pair prediction do not alone establish fusion benefit. He et al. 943, 949–50; supplementary note 6, supplementary fig. 12b.

Study and role	Data and research question	Approach	Findings and limits	Implication for model comparison
MIMIC Generative pretraining	Nucleotide and amino-acid tracks plus contextual inputs; reconstruction and prediction.	Use supplied sequence correspondence; reconstruct sequences from available input subsets.	mRNABench comparisons reuse published baseline scores with differing model inputs.	Assess reconstruction and prediction under their stated input conditions. Golkar et al., secs. 2, 4.1–4.2, app. D.3.
Structure-Aligned PLM Structural supervision	Sequence plus structure during training; contact prediction from sequence.	Residue-level contrastive learning and structure-token prediction.	Improves contact prediction in the tested setup; sequence alone is used at inference.	Extra training supervision differs from extra inference inputs. Chen et al., secs. 2.3, 3.1, 4.3.1, table 1.
OmniGenome Structural supervision	RNA sequence and secondary-structure labels; structure prediction and RNA design.	Predict structure from sequence and hidden nucleotides from sequence/structure context.	Training structures are predicted; design results include a genetic-algorithm search.	Assess the source of labels and the full prediction or design pipeline. Yang et al. 21930–33.
Coupled Mamba General fusion	Multiple modalities, including expression and methylation; classification.	Coupled state-space streams exchange hidden information.	The breast-cancer task uses multi-omics measurements, not DNA/RNA/protein sequences.	Compare evidence under task-specific conditions; re-evaluated by Fusion or Confusion. Li et al., sec. 3.2, eqs. 5–6, table 5; Rheude et al., sec. 4.2.
MSAlign Task-defined alignment	Measured spectra and candidate molecular structures; molecular identification.	Contrastive projections over frozen encoders, using candidate-based negatives.	Retrieval is limited to the candidate list; optional formula-filtered variants use the true molecular formula.	Judge alignment by whether it helps identify the correct molecule. Krzakala et al., secs. 2–3, 5.2–5.3.

Study and role	Data and research question	Approach	Findings and limits	Implication for model comparison
Comparative evaluations, benchmarks, and analyses				
Fusion or Confusion Architecture evaluation	Nineteen methods across nine datasets; does complex fusion improve prediction?	Reimplement methods and compare them with simple fusion and single-modality baselines.	No consistent aggregate complexity advantage; excludes some approaches and training settings.	Use strong baselines and retain the evaluation's stated scope. Rheude et al., secs. 4.2, 5.1.
COMET Biological benchmark	Seventeen tasks involving individual molecules and molecular pairs; model capabilities.	Evaluate across a biological task suite using model-specific adaptation protocols.	Task breadth permits varied tests; adaptation differs across evaluated models.	Separate representation quality from how each model is adapted. Ren et al., table 1, sec. 5.1, app. A.7.
mRNABench Representation benchmark	Mature-mRNA properties and functions; accessible information and generalization.	Compare linear probes on frozen, mean-pooled representations across data splits.	Performance and split effects vary by task; probing does not test full fine-tuning capability.	Match the readout and split to the intended biological use. Shi et al., secs. 4.2–4.3, 5.3, app. C.
Representation Alignment Analytical study	Synthetic and vision/language settings; relation between alignment and performance.	Analyze shared and unique task information using CKA and mutual KNN.	Higher measured alignment does not consistently correspond to better performance.	Motivates biological tests without providing biological validation. Tjandrasuwita et al., secs. 2–3.1, 5–6.

Note. Groups identify each study's principal role in this review, not exclusive study types. Model papers also include evaluations. Results across different tasks and input conditions are not a common performance ranking.

Evaluation and methodological choice

Table 1 brings together studies that change biological inputs, encoders, fusion operations, or training objectives. Their results require identifying what each experiment changes and holds fixed, and whether it evaluates representation relationships or task performance. BioLangFusion's RNA-stability task and IsoFormer's transcript-abundance task both use Spearman correlation, but their different targets and datasets prevent direct ranking by score (Mollaysa et al., sec. 3, table 1; Garau-Luis et al., secs. 4–5).

BioLangFusion's RNA-stability task provides a comparison of three fusion methods on the same biological target. The paper evaluates predictions using Spearman correlation, which measures how closely predicted scores preserve the ordering of measured stability. Higher correlation indicates that relatively stable RNAs tend to receive higher predicted scores than less stable RNAs; it does not establish that predictions have the correct numerical values. BioLangFusion's Table 1 reports correlations of 0.553 for RNA-FM alone, 0.539 for concatenation, 0.563 for MIL + Entropy, and 0.550 for cross-attention. Combining the views therefore does not improve this target under every tested method, and the table alone does not establish the statistical reliability of the differences (Mollaysa et al., sec. 3, table 1).

Interpreting these results also requires separating the fusion operation from the prediction head used with it. Section 3 states that all fusion methods share the same TextCNN prediction head, but Appendix A.4 lists 1,280 convolutional output channels for MIL + Entropy and 100 for cross-attention. Channel count specifies how many learned output features the convolution computes, so different counts would introduce an architectural difference beyond fusion. If these settings were used, performance differences could reflect either component or their interaction. The inconsistent descriptions leave it unclear whether the reported comparison held the head

configuration constant; they do not establish that the executed implementations differed (Mollaysa et al., sec. 3, app. A.4, lists 1–2).

[Implementation verification: Check BioLangFusion's author-released code and experiment configurations, if available, to clarify sequence preprocessing, DNA upsampling settings, and TextCNN channel counts. Identify the repository version and its connection to the reported experiments before using it to resolve differences in the paper's descriptions.]

IsoFormer first compares additional sequence inputs for predicting transcript abundance across tissues, using R^2 alongside Spearman correlation. R^2 compares squared prediction errors with those from always predicting the mean observed target value. With Nucleotide Transformer as the DNA and RNA encoder, Table 2 reports mean R^2 values of 0.36 ± 0.03 for RNA alone, 0.39 ± 0.01 for DNA plus RNA, and 0.43 ± 0.01 for DNA, RNA, and protein together. Table 2 reports results across five random seeds but does not explicitly define its ± notation. Table 3 identifies standard deviations across five seeds. The seeds vary the validation subset and initialization of non-pretrained parameters while retaining the same test set. The three-input configuration performs better in this comparison, supporting a benefit from the additional representations under the tested training setup (Garau-Luis et al., secs. 4.4–5.1, tables 2–3, app. B).

A second comparison changes the DNA encoder while retaining the three types of input. Enformer is a pretrained DNA-sequence model trained to predict regulatory measurements, including gene expression; IsoFormer uses it as an alternative DNA encoder. Replacing Nucleotide Transformer with Enformer raises mean R^2 from 0.43 ± 0.01 to 0.53 ± 0.01 (Garau-Luis et al., sec. 4.3, table 3). Enformer also accepts a much longer DNA context, approximately 196,000 nucleotides rather than 12,000. The longer interval can supply surrounding regulatory

sequence outside the shorter input, connecting this computational choice to the genomic context discussed earlier. The result concerns the encoder and its available sequence information together; it does not isolate the contributions of pretraining, context length, or fusion (Garau-Luis et al., secs. 4.3, 5.2, app. B, table 6).

The third comparison examines fusion more directly by replacing the aggregation module. These experiments use Enformer as the DNA encoder and give each alternative the same hyperparameter-search budget. One component, the Perceiver Resampler, uses attention to compress input representations into a fixed number of learned summary vectors. The Perceiver Resampler + cross-attention alternative compresses each modality separately before exchanging information through cross-attention. Linear projection + Perceiver Resampler instead projects embeddings to a common width, concatenates the modality tokens, and then summarizes the combined sequence. In both cases, the summary vectors represent information from the input rather than newly selected biological bases. C-Abstractor is a convolution-based summarizing module that uses adaptive mean pooling between one-dimensional residual-convolution blocks to preserve local sequence information. These alternatives test different ways to summarize representations and combine their information (Garau-Luis et al., sec. 5.3, fig. 4).

Under this comparison, both C-Abstractor and the cross-attention module achieve a reported mean R^2 of 0.53 ± 0.01. Their equal rounded results do not establish statistical equivalence, but they also do not demonstrate that cross-attention is uniquely best among the tested methods. This comparison evaluates alternative aggregation architectures under the stated training and search conditions while retaining the three input types. It therefore asks a different question from the modality ablation, which changes which inputs are supplied (Garau-Luis et al., sec. 5.3, table 4).

The sequence–structure studies test whether structural training makes useful information accessible from sequence representations. The Structure-Aligned Protein Language Model study evaluates contact prediction on selected proteins from CASP16, a protein-structure assessment, with sequence length limited to 512 amino-acid residues. It trains a prediction head while keeping each evaluated encoder frozen. Eligible residue pairs must be separated by more than six positions along the sequence. Here, L denotes the number of amino-acid residues in the protein sequence. Precision among the top L/5 predictions is the fraction of correct contacts among the L/5 highest-scoring eligible pairs. The reported precision increases from 0.181 for ESM2 650M to its structure-trained counterpart, SaESM2, at 0.288. This result supports improved access to residue-contact information through the frozen sequence encoder and trained prediction head (Chen et al., secs. 4.2–4.3.1, table 1).

OmniGenome's RNA-design task uses EternaV2, a benchmark containing 100 target secondary structures. Each puzzle asks for a sequence whose predicted secondary structure matches its target. The reported result of 74 solved puzzles belongs to OmniGenome186M+, the variant using structural context, operating within the genetic-algorithm search. This evaluates the combined model and search procedure on computational design puzzles. Matching a predicted structure does not experimentally demonstrate that the designed RNA has a desired cellular function (Yang et al. 21933, table 2).

MSAlign evaluates an alignment objective through a different task: retrieving the correct molecule from a candidate list for a measured spectrum. Its evaluation uses the MassSpecGym Formula split. The benchmark pairs measured mass spectra with known molecular identities and provides candidate lists for retrieval. With frozen spectrum and molecule encoders, MSAlign compares training losses for the projections that connect their representations. Candidate-based

contrastive learning, which trains against alternatives from the retrieval candidate list, outperforms in-batch negatives and regression alternatives in this comparison. This links the choice of alignment objective to retrieval performance under the tested conditions, rather than inferring usefulness from representation similarity alone (Krzakala et al., secs. 5.1, 5.3, table 5).

The need for controlled comparisons also extends to architectures evaluated on other kinds of biological input. Coupled Mamba offers an alternative to transformer fusion by coupling state-space streams. Its published evaluation includes classification of breast-cancer subtypes using mRNA expression, DNA methylation, and miRNA expression. As discussed in the biological background, these inputs describe measured RNA abundances and DNA modification states, whereas BioLangFusion and IsoFormer receive molecular sequences. The result supports a multi-omics application without settling a sequence-fusion choice (Li et al., secs. 3.2, 4.1, eqs. 5–6, table 5). Fusion or Confusion includes Coupled Mamba in a broader comparison, allowing its reported architectural benefits to be examined under another evaluation protocol (Rheude et al., sec. 4.2).

[Conditional extension: Explain how Coupled Mamba's state-space operation differs from attention if its architecture becomes a central comparison.]

Fusion or Confusion reimplements nineteen methods across nine datasets, comparing them with a baseline comprising modality encoders, projections, and a transformer head. Its simplicity is relative to the other designs, not an absence of attention. The study uses the same encoder architectures for each dataset across methods and standardized evaluation, finding no consistent aggregate advantage for greater complexity. Tuning begins from baseline-derived settings, which the authors acknowledge could favor their baseline (Rheude et al., secs. 3.1–3.3, 4.2, 5.1). These results qualify how broadly individual findings such as Coupled Mamba's can be

generalized. The study excludes early fusion, large-scale pretraining, and contrastive methods; its conclusion cannot be extended to LucaOne, MIMIC, or MSAlign as though their central training approaches had been tested (Rheude et al., sec. 5.1).

COMET and mRNABench organize biological evaluation at different levels. COMET spans seventeen tasks involving individual molecular sequences and molecular pairs, with both frozen-encoder and fine-tuned comparisons. In the fine-tuned comparisons, LucaOne uses low-rank adaptation (LoRA), which keeps pretrained weight matrices fixed and learns their adjustments through smaller trainable matrices. The other pretrained models receive full tuning that updates pretrained parameters directly. These results can therefore reflect both the representations and how they are adapted; the protocol alone does not establish which model benefits most from that difference (Ren et al., table 1, sec. 5.1, table 3, app. A.7).

mRNABench concentrates on mature mRNA tasks, including RNA half-life, localization, and mean ribosome load, the average number of ribosomes associated with an mRNA. Its linear probes leave the encoder fixed while a small prediction model learns a linear mapping from pooled embeddings to a task's numerical target or class scores. The embeddings are formed by averaging representations across sequence positions. This evaluates how readily a simple readout can use the pretrained representation, rather than the full capability a model might acquire through fine-tuning (Shi et al., secs. 3.1, 4.3, app. C).

The data split also determines what generalization the probe is asked to demonstrate. Related sequences can occur in both training and test sets under random splitting. mRNABench compares this with splits that separate sequences by chromosome, short sequence-pattern similarity, or evolutionary relationships between genes. No single strategy produces the largest performance reduction across every task, so the choice depends on which kinds of unfamiliar

examples the application requires (Shi et al., secs. 4.2, 5.3). Its compositional test examines a further distinction: models can encounter individual regulatory features during training yet perform poorly when tested on a previously unseen combination of those features. This probes the use of familiar elements in a new combination, beyond performance on ordinary held-out examples (Shi et al., sec. 5.4).

IsoFormer provides a related example by separating its test set by gene, keeping transcripts from the same gene on one side of the training–test boundary (Garau-Luis et al., sec. 4.4). This prevents sharing gene identities across that boundary, but gene-identity separation does not by itself ensure that evolutionarily related genes are separated. Its split and mRNABench's homology-based split therefore test different degrees of sequence relatedness (Shi et al., sec. 4.2).

In the reported scores in Table S11, MIMIC exceeds Evo 2 on four of seven tasks and Orthrus on four, although these are different task subsets. Evo 2 is a DNA foundation model, while Orthrus is a mature-RNA foundation model (Golkar et al., table S11; Shi et al., app. A). The reported Orthrus scores match mRNABench's Orthrus+MLM variant, which combines contrastive learning that brings related transcript representations closer with masked-token prediction that recovers hidden sequence units (Golkar et al., table S11; Shi et al., secs. 4.1, 5.1, tables 5–6). The comparator therefore already uses two complementary training objectives. MIMIC's advantage on some tasks compares models with different pretraining strategies as well as different available inputs; it cannot be attributed to combining biological inputs alone.

The MIMIC comparison also changes what information and prediction procedures are available. Coding-transcript configurations can include protein and codon inputs alongside nucleotide sequence. Within a configuration, MIMIC averages probe predictions across input

subsets, then selects the configuration for each task using validation data. Its baseline scores are reused from the published benchmark rather than recomputed. The results consequently compare complete prediction procedures with different available inputs; they do not isolate the contribution of fusion independently of the available inputs and prediction procedure (Golkar et al., sec. 4.2, app. D.3).

The evaluations distinguish several contributions to prediction. Removing an input tests its value under a specified training setup; replacing an encoder changes the available representations; changing an objective tests a learning rule. Attributing improvement to alignment additionally requires connecting the particular relationship changed or measured to task performance. An alignment score associated with better predictions provides observational evidence. Changing the model or objective to encourage that relationship tests whether the intervention improves prediction. This distinction determines what evidence can support a methodological choice based on alignment.

Summary

The reviewed studies can be assessed through three questions: What biological relationship connects the information? How does the model represent, establish, or measure that relationship? What evidence shows whether that relationship helps the specified biological task? The first question requires tracing what each input contains. Genomic context can supply regulatory sequence absent from a processed transcript, and translation removes nucleotide distinctions retained in the coding sequence (Garau-Luis et al., sec. 4.4; Nelson et al., sec. 27.1). Measurements of RNA abundance and DNA methylation describe molecular states (Li et al., sec. 4.1, table 5). These differences affect how correspondence between the inputs can be interpreted.

The second question distinguishes relationships supplied to a model from those learned or measured. BioLangFusion assumes coding correspondence and maps features to corresponding codon and amino-acid positions before fusion (Mollaysa et al., sec. 2.1). LucaOne's convergence analysis measures shared neighborhood relationships between nucleic-acid and protein embeddings associated with the same genes (He et al. 950, supplementary note 6 and supplementary fig. 12b). Structural supervision and molecular retrieval encourage specified pairings through training objectives (Chen et al., sec. 3.1; Krzakala et al., sec. 3). Mapping positions, measuring neighborhood similarity, and retrieving a molecule consequently answer different questions about correspondence. Their results must be interpreted according to what was supplied, learned, or tested.

The third question asks whether those relationships contribute to the biological task. The reviewed evaluations support benefits from additional inputs or structural supervision in particular settings (Mollaysa et al., table 1; Garau-Luis et al., tables 2–4; Chen et al., table 1). Characterizing correspondence alone does not establish a fusion benefit. Comparisons also

depend on adaptation procedures and data splits (Ren et al., sec. 5.1; Shi et al., secs. 4.3, 5.3). Where inputs, pretraining, and prediction procedures change together, improved performance supports the complete tested system but does not isolate the contribution of alignment (Golkar et al., sec. 4.2, app. D.3). The analytical alignment study finds that greater similarity does not consistently accompany better performance in its synthetic and nonbiological experiments. For biological fusion, this motivates testing alignment's usefulness while preserving task-relevant differences between inputs; it does not establish the outcome of that test (Tjandrasuwita et al., secs. 5–6).

Together, these questions connect the interpretation of alignment to choosing biological inputs, encoders, and combination methods for a defined task. Evidence that representations preserve a biological relationship can motivate a candidate approach; comparisons with strong individual encoders and simple fusion under consistent evaluation conditions can assess its predictive value (Rheude et al., sec. 5.3; Shi et al., secs. 4.2–4.3, 5.3). The biological claim determines what further validation is required: predicting abundance does not identify whether synthesis or degradation changed, and computationally matching a target structure does not demonstrate cellular function (Nelson et al., sec. 26.2; Yang et al. 21933). For the research considered here, the next question is which relationships can guide model selection for the intended biological task while retaining useful information specific to each input.

Works Cited

Alberts, Bruce, et al. *Essential Cell Biology*. 6th ed., e-book ed., W. W. Norton, 2023.

Alberts, Bruce, et al. *Molecular Biology of the Cell*. 7th ed., W. W. Norton, 2022.

Chen, Can (Sam), et al. "Structure-Aligned Protein Language Model." *arXiv*, version 2, 17 Dec. 2025, https://arxiv.org/abs/2505.16896v2.

Garau-Luis, Juan Jose, et al. "Multi-modal Transfer Learning between Biological Foundation Models." *Advances in Neural Information Processing Systems*, vol. 37, 2024, pp. 78431–50, https://doi.org/10.52202/079017-2493.

Golkar, Siavash, et al. "MIMIC: A Generative Multimodal Foundation Model for Biomolecules." *arXiv*, version 1, 27 Apr. 2026, https://arxiv.org/abs/2604.24506v1.

He, Yong, et al. "Generalized Biological Foundation Model with Unified Nucleic Acid and Protein Language." *Nature Machine Intelligence*, vol. 7, 2025, pp. 942–53, https://doi.org/10.1038/s42256-025-01044-4.

Ji, Yanrong, et al. "DNABERT: Pre-Trained Bidirectional Encoder Representations from Transformers Model for DNA-Language in Genome." *Bioinformatics*, vol. 37, no. 15, 2021, pp. 2112–20, https://doi.org/10.1093/bioinformatics/btab083.

Krzakala, Paul, et al. "MSAlign: Aligning Molecule and Mass Spectra Representations for Metabolite Identification." *arXiv*, version 2, 25 Sept. 2026, https://arxiv.org/abs/2605.19752v2.

Li, Wenbing, et al. "Coupled Mamba: Enhanced Multi-modal Fusion with Coupled State Space Model." *Advances in Neural Information Processing Systems*, vol. 37, 2024, pp. 59808–32, https://doi.org/10.52202/079017-1910.

Mollaysa, Amina, et al. “BioLangFusion: Multimodal Fusion of DNA, mRNA, and Protein Language Models.” *arXiv*, version 1, 10 June 2025, https://arxiv.org/abs/2506.08936v1.

Nelson, David L., Michael M. Cox, and Aaron A. Hoskins. *Lehninger Principles of Biochemistry*. 8th ed., e-book ed., Macmillan Learning, 2021.

Ota, Nobuyuki. “Central Dogma Transformer: Towards Mechanism-Oriented AI for Cellular Understanding.” *arXiv*, version 2, 10 Jan. 2026, https://arxiv.org/abs/2601.01089v2.

Ren, Yuchen, et al. “COMET: Benchmark for Comprehensive Biological Multi-omics Evaluation Tasks and Language Models.” *arXiv*, version 1, 13 Dec. 2024, https://arxiv.org/abs/2412.10347v1.

Rheude, Tillmann, Roland Eils, and Benjamin Wild. “Fusion or Confusion? Multimodal Complexity Is Not All You Need.” *arXiv*, version 3, 7 May 2026, https://arxiv.org/abs/2512.22991v3.

Shi, Ruian (Ian), et al. “mRNABench: A Curated Benchmark for Mature mRNA Property and Function Prediction.” *bioRxiv*, version 1, 8 July 2025, https://www.biorxiv.org/content/10.1101/2025.07.05.662870v1.

Tjandrasuwita, Megan, et al. “Understanding the Emergence of Multimodal Representation Alignment.” *Proceedings of the 42nd International Conference on Machine Learning*, vol. 267, PMLR, 2025, pp. 59723–60, https://proceedings.mlr.press/v267/tjandrasuwita25a.html.

Yang, Heng, et al. “Bridging Sequence-Structure Alignment in RNA Foundation Models.” *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 39, no. 20, 2025, pp. 21929–37, https://doi.org/10.1609/aaai.v39i20.35500.

AI Disclosure

Chase Di Maria Breisinger directed this review through dialogue with Codex (OpenAI). He supplied original and AI-refined Obsidian notes, three biology textbooks, the research canvas, websites and course links, an annotated submission, and research-planning guidance. Research-group recommendations and his additions shaped the selected papers; he relayed the 10 October meeting's emphasis on narrowing scope. By repeatedly reading and questioning generated passages, he challenged unclear terminology, unsupported implications, repetition, weak transitions, and citation locators. His feedback shaped explanations, section organization, and the connection between alignment and model selection. He requested plans before substantial changes and approved successive revisions.

Julian McKinley contributed earlier draft material prepared with Claude (Anthropic), whose disclosed assistance included locating sources, drafting, formatting, and checking publication information. That material helped direct attention to differences among genomic context, processed transcripts, coding sequences, and translated proteins. The available record does not identify which individual explanations originated in Julian's prior knowledge, his prompts, or Claude's output, or which sources he independently verified. Useful biological explanations from that material were retained where supported by the textbooks.

Codex generated most of the prose, comparative table, and synthesis. It explained biological and computational concepts, compared methods and evaluations, and handled citations, formatting, and PDF production. Some generated wording and source attributions required correction. The broad-to-specific Introduction and the Summary's three questions developed through Chase's feedback, with the research canvas supplying project context rather

than scientific evidence. Additional Codex agents checked source support, prose, and layout. These were AI-assisted checks, not independent human reviews.

The scientific Works Cited contains fourteen research papers and three biology books. Textbooks ground established biology; papers support method-specific and experimental claims. Codex checked citation occurrences and the comparative table against the retained sources, qualifying claims where descriptions were ambiguous or inconsistent. Source selection was not systematic or exhaustive. Printed textbook pages are used where stable, and reflowed editions are cited by chapter or section. The scientific discussion excludes local project results. The implementation-verification note identifies unresolved questions for future checks of author code and experiment configurations. It does not report verified implementation behavior.

Chase's close review and approval of revisions are documented; complete independent verification of final claims and references by both authors is not. No experiments or notebooks were rerun. The ethical implications of research automation have not been fully examined. The review remains subject to the authors' correction. This rendition is dated 12 October 2026.

Preparation guidance: Leedy, Paul D., and Jeanne Ellis Ormrod, with Laura Ruth Johnson. *Practical Research: Planning and Design*. 12th ed., Global ed., Pearson, 2021, pp. 98–102. Chapter 3's ten guidelines informed organization, relevance, transitions, attribution, synthesis, revision, and feedback. Chase applied this guidance to the advance organizer, section conclusions, and final Summary. He adapted point 9's bold red placeholders (pp. 101–102) to identify biological clarifications, a worked computational example, implementation verification, and conditional extensions. These annotations identify possible further work. The book guides preparation rather than scientific claims.